

1 Integral Notation

Partially evaluating a conditional probability distribution on any $y \in Y$ yields a probability distribution that concentrates on the level set $f^{-1}(y)$,

$$\begin{aligned}\pi_y^f : \mathcal{X} &\rightarrow [0, 1] \\ x &\mapsto \pi^f(x \mid y).\end{aligned}$$

When paired with an integrand $g : X \rightarrow \mathbb{R}$ this collection of probability distributions defines a **conditional expectation function**,

$$\begin{aligned}e_g : Y &\rightarrow \mathbb{R} \\ y &\mapsto \mathbb{E}_{\pi_y^f}[g].\end{aligned}$$

The law of total expectation ensures that the pushforward expectation of this conditional expectation function is always equal to the original expectation value,

$$\mathbb{E}_\pi[g] = \mathbb{E}_{f_*\pi}[e_g].$$

Ideally we'd be able to write the law of total expectation more compactly, packing everything into one line. Unfortunately any compact notation is frustrated by the standard notational convention of ignoring arguments when denoting expectation values. For example a notation that replaces y with a placeholder “.”,

$$\mathbb{E}_\pi[g] = \mathbb{E}_{f_*\pi}[e_g] = \mathbb{E}_{f_*\pi}[\mathbb{E}_{\pi_y^f}[g]],$$

is not only hard to read but can be ambiguous in applications where there are multiple spaces on which f and g might act.

At the same time a more explicit notation like

$$\mathbb{E}_\pi[g] = \mathbb{E}_{f_*\pi}[e_g] = \mathbb{E}_{f_*\pi}[\mathbb{E}_{\pi_y^f}[g(x)]]$$

breaks the standard notational convention without clarifying on what spaces the probability distributions $f_*\pi$ and π_y^f are defined.

Conditional probability theory is one place where the more explicit integral notation for expectation values that we first discussed in **previous chapter** becomes incredibly useful. To make the notation as compact as possible we will need to introduce a *conditional variable* z_y that takes values in only the level set corresponding to the output point $y \in Y$,

$$z_y \in f^{-1}(y) \subset X.$$

Conditional variables allow us to decompose any input variable into an output variable and a corresponding conditional variable,

$$x = (y, z_y).$$

Note that the right hand-side isn't an ordered pair because the possible values of the second variable will in general depend on the choice of the first variable. For the mathematically-curious this construction is known as a **semi-direct product**.

Using conditional variables we can write the conditional expectation value as

$$\begin{aligned} e_g(y) &= \mathbb{E}_{\pi_y^f}[g] \\ &= \int \pi^f(dx \mid y) g(x), \end{aligned}$$

with the law of total expectation becoming

$$\begin{aligned} \mathbb{E}_\pi[g] &= \mathbb{E}_{f_*\pi}[e_g] \\ \int \pi(dx) g(x) &= \int f_*\pi(dy) e_g(y) \\ \int \pi(dx) g(x) &= \int f_*\pi(dy) \int \pi^f(dx \mid y) g(x). \end{aligned}$$

Convert $dx \mid y$ **to** $dx_y \mid y$.

I will make extensive use of this compact notation anytime conditional probability theory might come into play which, as we will see, will be often.

2 Conditional Probability Density Functions

2.1 Non-Null Partitions

Constructing conditional probability density functions for non-null partitions is relatively straightforward. Recall that in this case we can define a conditional probability as

$$\pi^{\mathcal{P}}(\mathbf{x} \mid \mathbf{c}_j) = \frac{\pi(\mathbf{x} \cap \mathbf{c}_j)}{\pi(\mathbf{c}_j)}.$$

More generally for any integrand $g : X \rightarrow \mathbb{R}$ these conditional probability allocations define a conditional expectation value,

$$\mathbb{E}_{\pi_{\mathbf{c}_j}^{\mathcal{P}}}[g] = \frac{\mathbb{E}_\pi[I_{\mathbf{c}_j} \cdot g]}{\mathbb{E}_\pi[I_{\mathbf{c}_j}]}.$$

Given a convenient reference measure μ we can convert these expectation values into μ -informed integrals with the introduction of a Radon-Nikodym derivative. On the left-hand side we have

$$\mathbb{E}_{\pi_{\mathbf{c}_j}^{\mathcal{P}}}[g] = \mathbb{I}_\mu \left[\frac{d\pi_{\mathbf{c}_j}^{\mathcal{P}}}{d\mu} \cdot g \right]$$

and on the right-hand side we have

$$\begin{aligned}\frac{\mathbb{E}_\pi[I_{c_j} \cdot g]}{\mathbb{E}_\pi[I_{c_j}]} &= \frac{\mathbb{I}_\mu\left[\frac{d\pi}{d\mu} \cdot I_{c_j} \cdot g\right]}{\mathbb{I}_\mu\left[\frac{d\pi}{d\mu} \cdot I_{c_j}\right]} \\ &= \mathbb{I}_\mu\left[\frac{\frac{d\pi}{d\mu} \cdot I_{c_j}}{\mathbb{I}_\mu\left[\frac{d\pi}{d\mu} \cdot I_{c_j}\right]} \cdot g\right].\end{aligned}$$

Consequently

$$\begin{aligned}\mathbb{E}_{\pi_{c_j}^{\mathcal{P}}}[g] &= \frac{\mathbb{E}_\pi[I_{c_j} \cdot g]}{\mathbb{E}_\pi[I_{c_j}]} \\ \mathbb{I}_\mu\left[\frac{d\pi_{c_j}^{\mathcal{P}}}{d\mu} \cdot g\right] &= \mathbb{I}_\mu\left[\frac{\frac{d\pi}{d\mu} \cdot I_{c_j}}{\mathbb{I}_\mu\left[\frac{d\pi}{d\mu} \cdot I_{c_j}\right]} \cdot g\right]\end{aligned}$$

which is true for any integrand $g : X \rightarrow \mathbb{R}$ if, and only if,

$$\frac{d\pi_{c_j}^{\mathcal{P}}}{d\mu} \stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu} \cdot I_{c_j}}{\mathbb{I}_\mu\left[\frac{d\pi}{d\mu} \cdot I_{c_j}\right]}.$$

Repeating this calculation for each partition cell c_j defines a family of probability density functions that completely specifies the conditional probability distribution in the context of the reference measure μ .

In words a conditional probability density function for a non-null partition is given by truncating the unconditional probability density function to each partition cell and then scaling by the reciprocal of the cell probability to ensure the proper normalization.

figure

2.2 The Problem with Null Partitions

Defining conditional probability density functions when the partition cells might be allocated vanishing probability is, unfortunately, a bit more complicated. Because we can no longer define conditional probabilities as a ratio of unconditional probabilities we need to consult the law of total expectation in order to define conditional probability density functions,

$$\int \pi(dx) g(x) = \int f_*\pi(dy) \int \pi^f(dx | y) g(x),$$

In order to move forwards we need to convert the inner expectation into a reference measure-informed integral. The subtlety, however, is exactly *how* we can integrate over the level sets.

Consider, for example, a reference measure λ over the entire input space for which each conditional probability distribution $\pi^f(x | y)$ is absolutely continuous. In theory we can write the expectation over any particular level set as a λ -informed integral using a Radon-Nikodym derivative,

$$\int \pi^f(dx | y) g(x) = \int \lambda(dx) \frac{d\pi^f}{d\lambda}(x | y) g(x).$$

Actually not mathematically valid. While this is mathematically valid the necessary Radon-Nikodym derivative $d\pi^f/d\lambda$ is somewhat weird object. For example remember that any subset that doesn't intersect with a given level set is allocated zero conditional probability,

$$\begin{aligned} 0 &= \int \pi^f(dx | y) I_x(x) \\ &= \int \lambda(dx) \frac{d\pi^f}{d\lambda}(x | y) I_x(x). \end{aligned}$$

This is true, however, if and only if

$$\frac{d\pi^f}{d\lambda}(x | y) \stackrel{\lambda, \nu}{=} 0.$$

In other words $d\pi^f/d\lambda(x | y)$ has to be effectively zero outside of the level set $f^{-1}(y)$.

At the same time if the level set is allocated vanishing reference measure,

$$\lambda(f^{-1}(y)) = 0$$

then any integrand that is non-zero only within the level set $f^{-1}(y)$ should give the same integral as an integrand that is identically zero within that level set. Consequently we should have

$$\begin{aligned} \int \pi^f(dx | y) g(x) &= \int \lambda(dx) \frac{d\pi^f}{d\lambda}(x | y) g(x) \\ &= \int \lambda(dx) 0 \cdot g(x) \\ &= 0 \end{aligned}$$

regardless of g . That vanishing integral then propagates through the law of total expectation to give

$$\begin{aligned} \int \pi(dx) g(x) &= \int f_*\pi(dy) \int \pi^f(dx | y) g(x) \\ &= \int f_*\pi(dy) 0 \\ &= 0 \end{aligned}$$

for all expectands g , which cannot hold for arbitrary probability distributions π .

In order to ensure non-trivial expectations the object $d\pi^f/d\lambda(x | y)$ cannot be a regular function. Instead it would have to be a generalized function that can inflate a null subset to have non-vanishing integrals, mirroring the construction of the Dirac delta “function” that we encountered in **previous chapter**.

In practice, however, it’s much more straightforward if we restrict the inner expectation in the law of total probability to particular level sets and define conditional probability density functions relative to a family of reference measures, each of which is restricted to a particular level set,

$$\int \pi^f(dx | y) g(x) = \int \eta_y(dx) \frac{d\pi^f}{d\eta_y}(x | y) g(x).$$

In order to relate this to probability density functions defined over the input and output spaces we’ll need to learn how to disintegrate measures.

2.3 Disintegrating Measures

Disintegrating measures, however, is more subtle than disintegrating probability distributions.

Firstly in order to avoid the kinds of ambiguous results that arise when we allow infinities into calculations we need to restrict our consideration to measures that are σ -finite. Moreover when generalizing the law of total expectation into a law of total integration we cannot rely on the pushforward of an initial σ -finite measure because it will not always be σ -finite.

For example consider a rigid two-dimensional real space $\mathbb{R}^2$ equipped with the two-dimensional Lebesgue measure λ^2 and a projection function

$$\begin{aligned} \varpi_1 : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x_1, x_2) &\mapsto x_1. \end{aligned}$$

The Lebesgue measure λ_2 is σ -finite, allocating finite measure to every measurable subset that can be encapsulated in a finite rectangle: if

$$x \subset [0, 1] \times [0, 1]$$

then

$$\begin{aligned} \lambda^2(x) &< \lambda^2([0, 1] \times [0, 1]) \\ &< l([0, 1]) \cdot l([0, 1]) \\ &< 1 \cdot 1 \\ &< 1. \end{aligned}$$

Pushing λ^2 forward along ϖ_1 , however, results in a measure that allocates infinite measure to finite intervals,

$$\begin{aligned}
(\varpi_1)_*\lambda^2([0, 1]) &= \lambda^2(\varpi_1^*[0, 1]) \\
&= \lambda^2([0, 1] \times (\infty, \infty)) \\
&= l([0, 1]) \cdot l((-\infty, \infty)) \\
&= 1 \cdot \infty \\
&= \infty.
\end{aligned}$$

figure

In order to define a well-behaved law of total integration for σ -finite measures we need to specify not only a measurable function $f : X \rightarrow Y$ but also a σ -finite reference measure over the output space Y . Formally given

1. an input measurable space $(X, \mathcal{X})$,
2. a σ -finite measure $\mu : \mathcal{X} \rightarrow [0, \infty]$,
3. an output measurable space $(Y, \mathcal{Y})$ where every atomic subset is measurable,
4. a $(\mathcal{X}, \mathcal{Y})$ -measurable function $f : X \rightarrow Y$,
5. and finally a σ -finite measure $\nu : \mathcal{Y} \rightarrow [0, \infty]$

there exists at least one conditional measure

$$\begin{aligned}
\mu^{f, \nu} : \mathcal{X} \times Y &\rightarrow [0, \infty] \\
x, y &\mapsto \mu^f(x \mid y)
\end{aligned}$$

that defines a $(\mathcal{Y}, \mathcal{B}_{\mathbb{R}})$ -measurable function when partially evaluated on the first argument,

$$\begin{aligned}
\mu_x^{f, \nu} : Y &\rightarrow [0, \infty] \\
y &\mapsto \mu^{f, \nu}(x \mid y)
\end{aligned}$$

for all $x \in \mathcal{X}$, and a σ -finite measure when partially evaluated on the second argument,

$$\begin{aligned}
\mu_{y, \nu}^f : \mathcal{X} &\rightarrow [0, \infty] \\
x &\mapsto \mu^{f, \nu}(x \mid y)
\end{aligned}$$

ν -almost all $y \in Y$. Any conditional measure almost always concentrates on a given level set,

$$\mu_y^{f, \nu}(x) \stackrel{\nu}{=} 0$$

for any $x \in \mathcal{X}$ with $x \cap f^{-1}(y)$ while also satisfying a law of total integration,

$$\int \mu(dx) g(x) = \int \nu(dy) \int \mu^{f,\nu}(dx_y | y) g(x),$$

for any well-behaved integrand $g : X \rightarrow \mathbb{R}$.

If $f_*\mu$ is σ -finite then we can always take $\nu = f_*\mu$ so that the law of total integration mirrors the law of total expectation. This is always possible if μ is a finite measure, with $\mu(X) < \infty$, but not always viable when μ is only σ -finite. In particular we have to be careful when disintegrating Lebesgue and counting measures!

2.4 General Implicit Partitions

Armed with a technique for disintegrating measures we are now equipped to construct conditional probability density functions for any conditional probability distribution.

Let's consider our general conditioning set up,

1. an input measurable space $(X, \mathcal{X})$,
2. a probability distribution $\pi : \mathcal{X} \rightarrow [0, 1]$,
3. an output measurable space $(Y, \mathcal{Y})$ where every atomic subset is measurable,
4. and a $(\mathcal{X}, \mathcal{Y})$ -measurable function $f : X \rightarrow Y$,

and introduce convenient σ -finite reference measures for **mention absolute continuity requirements!**

1. the input space $\mu : \mathcal{X} \rightarrow [0, \infty]$,
2. the output space $\nu : \mathcal{Y} \rightarrow [0, \infty]$,
3. and each level set, $\eta_y : \mathcal{F}_y \rightarrow [0, \infty]$.

Starting with the law of total expectation,

$$\int \pi(dx) g(x) = \int f_*\pi(dy) \int \pi^f(dx | y) g(x),$$

our goal will be to convert all of the expectations into iterated integrals informed by ν and the η_y .

Let's start with the left-hand side,

$$\int \pi(dx) g(x) = \int \mu(dx) \frac{d\pi}{d\mu}(x) g(x).$$

At this point we can disintegrate μ with respect to f and ν ,

$$\begin{aligned}\int \pi(\mathrm{d}x) g(x) &= \int \mu(\mathrm{d}x) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) g(x) \\ &= \int \nu(\mathrm{d}y) \int \mu^{f,\nu}(\mathrm{d}x | y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) g(x).\end{aligned}$$

Incorporating our reference measures over the level sets then gives

$$\begin{aligned}\int \pi(\mathrm{d}x) g(x) &= \int \nu(\mathrm{d}y) \int \mu^{f,\nu}(\mathrm{d}x | y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) g(x) \\ &= \int \nu(\mathrm{d}y) \int \eta_y(\mathrm{d}x) \frac{\mathrm{d}\mu^{f,\nu}}{\mathrm{d}\eta_y}(x | y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) g(x) \\ &= \int \nu(\mathrm{d}y) \int \eta_y(\mathrm{d}x) \left[\frac{\mathrm{d}\mu^{f,\nu}}{\mathrm{d}\eta_y}(x | y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) \right] g(x).\end{aligned}$$

Similarly for the right-hand side we have

$$\begin{aligned}\int f_*\pi(\mathrm{d}y) \int \pi^f(\mathrm{d}x | y) g(x) &= \int \nu(\mathrm{d}y) \frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \int \pi^f(\mathrm{d}x | y) g(x) \\ &= \int \nu(\mathrm{d}y) \frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \int \eta_y(\mathrm{d}x) \frac{\mathrm{d}\pi^f}{\mathrm{d}\eta_y}(x | y) g(x).\end{aligned}$$

Because $\mathrm{d}f_*\pi/\mathrm{d}\nu$ is a constant with respect to the inner integral we can pull it in to give

$$\begin{aligned}\int f_*\pi(\mathrm{d}y) \int \pi^f(\mathrm{d}x | y) g(x) &= \int \nu(\mathrm{d}y) \frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \int \eta_y(\mathrm{d}x) \frac{\mathrm{d}\pi^f}{\mathrm{d}\eta_y}(x | y) g(x) \\ &= \int \nu(\mathrm{d}y) \int \eta_y(\mathrm{d}x) \frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \frac{\mathrm{d}\pi^f}{\mathrm{d}\eta_y}(x | y) g(x) \\ &= \int \nu(\mathrm{d}y) \int \eta_y(\mathrm{d}x) \left[\frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \frac{\mathrm{d}\pi^f}{\mathrm{d}\eta_y}(x | y) \right] g(x).\end{aligned}$$

Combining these results gives

$$\begin{aligned}\int \pi(\mathrm{d}x) g(x) &= \int f_*\pi(\mathrm{d}y) \int \pi^f(\mathrm{d}x | y) g(x) \\ \int \nu(\mathrm{d}y) \int \eta_y(\mathrm{d}x) \left[\frac{\mathrm{d}\mu^{f,\nu}}{\mathrm{d}\eta_y}(x | y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) \right] g(x) &= \int \nu(\mathrm{d}y) \int \eta_y(\mathrm{d}x) \left[\frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \frac{\mathrm{d}\pi^f}{\mathrm{d}\eta_y}(x | y) \right] g(x)\end{aligned}$$

This is true for all integrands $g : X \rightarrow \mathbb{R}$ if and only if

$$\frac{\mathrm{d}\mu^{f,\nu}}{\mathrm{d}\eta_y}(x | y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) \stackrel{\nu, \eta_y}{=} \frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \frac{\mathrm{d}\pi^f}{\mathrm{d}\eta_y}(x | y).$$

whenever $f(x) = y$.

If we fix $y = f(x)$ then we can rearrange this condition to give

$$\begin{aligned} \frac{d\pi}{d\mu}(x) &\stackrel{\eta_{f(x)}}{=} \frac{df_*\pi}{d\nu}(f(x)) \frac{\frac{d\pi^f}{d\eta_{f(x)}}(x \mid f(x))}{\frac{d\mu^{f,\nu}}{d\eta_{f(x)}}(x \mid f(x))} \\ \frac{d\pi}{d\mu}(x) &\stackrel{\mu}{=} \frac{df_*\pi}{d\nu}(f(x)) \frac{d\pi^f}{d\mu^{f,\nu}}(x \mid f(x)). \end{aligned}$$

This is known as the **product rule** for probability density functions. The product rule allows us to construct either the ambient probability density function, the conditional probability density function, or the pushforward probability density function from knowledge of the other two. For example if we know the ambient probability density function and the pushforward probability density function then the conditional probability density function is given by

$$\frac{d\pi^f}{d\mu^{f,\nu}}(x \mid f(x)) \stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x)}{\frac{df_*\pi}{d\nu}(f(x))}.$$

Consider, for example, a function $f : X \rightarrow \mathbb{N}$ which induces a countable partition. The counting measure is a natural reference measure for the output space, $\nu = \chi$, and assuming that $f_*\mu(\{y\}) > 0$ for all $y \in \mathbb{N}$ we can take η_y to be μ restricted to each level set,

$$\eta_y = \mu^{f,\chi} = I_{f^{-1}(y)} \cdot \mu.$$

Applying the previous formula then gives **we can divide by indicator function only when $x \in f^{-1}(y)$ and indicator function doesn't return zero; there's always an**

implicit constraint

$$\begin{aligned}
\frac{d\pi^f}{d\mu^{f,\nu}}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x)}{\frac{df_*\pi}{d\nu}(f(x))} \\
\frac{d\pi^f}{d(I_{f^{-1}(y)} \cdot \mu)}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x)}{\frac{df_*\pi}{d\chi}(f(x))} \\
\frac{1}{I_{f^{-1}(y)}(x)} \frac{d\pi^f}{d\mu}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x)}{\frac{df_*\pi}{d\chi}(f(x))} \\
\frac{d\pi^f}{d\mu}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x) \cdot I_{f^{-1}(y)}(x)}{\frac{df_*\pi}{d\chi}(f(x))} \\
\frac{d\pi^f}{d\mu}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x) \cdot I_{f^{-1}(y)}(x)}{f_*\pi(\{f(x)\})} \\
\frac{d\pi^f}{d\mu}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x) \cdot I_{f^{-1}(y)}(x)}{\pi(f^*\{f(x)\})} \\
\frac{d\pi^f}{d\mu}(x \mid f(x)) &\stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x) \cdot I_{f^{-1}(y)}(x)}{\mathbb{I}_\mu \left[\frac{d\pi}{d\mu} \cdot I_{f^{-1}(f(x))} \right]}.
\end{aligned}$$

Denoting $f^{-1}(y) = c_y$ and

$$\frac{d\pi^f}{d\mu}(x \mid y) = \frac{d\pi^{\mathcal{P}}}{d\mu}(x \mid c_y)$$

this becomes

$$\frac{d\pi^{\mathcal{P}}}{d\mu}(x \mid c_y) \stackrel{\mu}{=} \frac{\frac{d\pi}{d\mu}(x) \cdot I_{c_y}(x)}{\mathbb{I}_\mu \left[\frac{d\pi}{d\mu} \cdot I_{c_y} \right]},$$

consistent with what we derived above.

3 Explicit Formula For Pushforward Probability Density Functions

Disintegrating a reference measure also gives us an explicit formula for pushforward probability density functions. We start with the definition of pullback expectation values: for sufficiently measurable $f : X \rightarrow Y$ and $h : Y \rightarrow \mathbb{R}$ we have

$$\mathbb{E}_\pi[h \circ f] = \mathbb{E}_{f_*\pi}[h],$$

or, equivalently,

$$\int \pi(\mathrm{d}x) h(f(x)) = \int f_*\pi(\mathrm{d}y) h(y).$$

Our goal will be to convert these expectations into equivalent integrals so that we can compare the integrands, at least up to null subsets. For example given a convenient reference measure ν over Y we can write the right-hand side as

$$\begin{aligned} \int f_*\pi(\mathrm{d}y) h(y) &= \int \nu(\mathrm{d}y) \frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) h(y) \\ &= \int \nu(\mathrm{d}y) \left[\frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \right] h(y). \end{aligned}$$

Similarly given a convenient reference measure μ over X we can write the left-hand side as

$$\int \pi(\mathrm{d}x) h(f(x)) = \int \mu(\mathrm{d}x) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) h(f(x)).$$

Disintegrating μ with respect to f and ν then gives

$$\begin{aligned} \int \pi(\mathrm{d}x) h(f(x)) &= \int \mu(\mathrm{d}x) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) h(f(x)) \\ &= \int \nu(\mathrm{d}y) \int \mu^{f,\nu}(\mathrm{d}x \mid y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) h(f(x)). \end{aligned}$$

Because the function $h \circ f : X \rightarrow \mathbb{R}$ yields the same output for any $x \in f^{-1}(y)$ it is a constant with respect to the inner integral that can be factored out,

$$\begin{aligned} \int \pi(\mathrm{d}x) h(f(x)) &= \int \nu(\mathrm{d}y) \int \mu^{f,\nu}(\mathrm{d}x \mid y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) h(f(x)) \\ &= \int \nu(\mathrm{d}y) \left[\int \mu^{f,\nu}(\mathrm{d}x \mid y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) \right] h(y). \end{aligned}$$

Combining the transformed left-hand and right-hand sides together gives

$$\begin{aligned} \int \pi(\mathrm{d}x) h(f(x)) &= \int f_*\pi(\mathrm{d}y) h(y) \\ \int \nu(\mathrm{d}y) \left[\int \mu^{f,\nu}(\mathrm{d}x \mid y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x) \right] h(y) &= \int \nu(\mathrm{d}y) \left[\frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \right] h(y). \end{aligned}$$

Because both sides of this equation are ν -informed integrals we have equality if and only if the integrands are equal up to ν -null subsets. In particular we have equality for any integrand $h : Y \rightarrow \mathbb{R}$ if and only if

$$\frac{\mathrm{d}f_*\pi}{\mathrm{d}\nu}(y) \stackrel{\nu}{=} \int \mu^{f,\nu}(\mathrm{d}x \mid y) \frac{\mathrm{d}\pi}{\mathrm{d}\mu}(x).$$

In theory this gives us an explicit formula for pushforward probability density functions. Implementing this result in practice, however, will not be straightforward unless we happen to have an explicit method to integrate the initial probability density function over each level set of f .

4 Simplifications For Product Spaces

Conditional probability theory can be intimidating in its full generality, but it becomes much more straightforward when applied to the natural projection functions of product spaces. Consider, for example, a product space consisting of I component spaces,

$$X^{1:I} = \times_{i=1}^I X_i,$$

where

$$1:I = \{1, \dots, I\}$$

denotes the component indices.

Given any subset of component indices

$$\mathbf{i} = \{i_1, \dots, i_J\} \subset 1:I$$

we can decompose any product variable

$$x_{1:I} = (x_1, \dots, x_I)$$

into a marginal variable

$$x_{\mathbf{i}} = (x_{i_1}, \dots, x_{i_J}) \in \times_{i \in \mathbf{i}} X_i$$

and a cross sectional variable

$$x_{\setminus \mathbf{i}} = (x_i \mid i \notin \mathbf{i}) \in \{x_i\} \times X^{\setminus \mathbf{i}},$$

where

$$\setminus \mathbf{i} = \{i \mid i \notin \mathbf{i}\}.$$

This decomposition is accompanied by the projection function

$$\varpi_{\mathbf{i}} : X^{1:I} \rightarrow X^{\mathbf{i}}$$

with the level sets

$$\varpi_{\mathbf{i}}^{-1}(x_{\mathbf{i}}) = \{x_{\mathbf{i}}\} \times X^{\setminus \mathbf{i}}.$$

Reference measures on each component space immediately define a product reference measure over the product space,

$$\nu^I = \times_{i=1}^I \nu_i.$$

At the same time the product measure

$$\nu^{\mathbf{i}} = \times_{i \in \mathbf{i}} \nu_i$$

defines a reference measure over the marginal space $X^{\mathbf{i}}$ which also happens to be the output space of the projection function $\varpi_{\mathbf{i}}$. Finally the product of the remaining component measures,

$$\nu^{\setminus \mathbf{i}} = \times_{i \in \setminus \mathbf{i}} \nu_i,$$

defines a reference measure for each cross section $\{x_{\mathbf{i}}\} \times X^{\setminus \mathbf{i}}$, and hence each level set of the projection function. In summary by combining the component reference measures together appropriately we can construct reference measures for the input space, output space, and level sets of the projection function $\varpi_{\mathbf{i}}!$

Product measure disintegrates into product measure over conditional components

$$(\nu^I)^{\varpi_{\mathbf{i}}, \nu^{\mathbf{i}}} = \nu^{\setminus \mathbf{i}}.$$

These reference measures allow us to construct conditional probability density functions over the cross sections that satisfy the product rule

$$\begin{aligned} \frac{d\pi}{d\nu^I}(x_{1:I}) &\stackrel{\nu^I}{=} \frac{d(\varpi_{\mathbf{i}})_*\pi}{d\nu^{\mathbf{i}}}(\varpi_{\mathbf{i}}(x_{1:I})) \frac{d\pi^{\varpi_{\mathbf{i}}}}{d(\nu^I)^{\varpi_{\mathbf{i}}, \nu^{\mathbf{i}}}}(x_{\setminus \mathbf{i}} \mid \varpi_{\mathbf{i}}(x_{1:I})) \\ \frac{d\pi}{d\nu^I}(x_{1:I}) &\stackrel{\nu^I}{=} \frac{d(\varpi_{\mathbf{i}})_*\pi}{d\nu^{\mathbf{i}}}(\varpi_{\mathbf{i}}(x_{1:I})) \frac{d\pi^{\varpi_{\mathbf{i}}}}{d\nu^{\setminus \mathbf{i}}}(x_{\setminus \mathbf{i}} \mid \varpi_{\mathbf{i}}(x_{1:I})) \\ \frac{d\pi}{d\nu^I}(x_{1:I}) &\stackrel{\nu^I}{=} \frac{d(\varpi_{\mathbf{i}})_*\pi}{d\nu^{\mathbf{i}}}(x_{\mathbf{i}}) \frac{d\pi^{\varpi_{\mathbf{i}}}}{d\nu^{\setminus \mathbf{i}}}(x_{\setminus \mathbf{i}} \mid x_{\mathbf{i}}). \end{aligned}$$

For example if $X^{1:I} = \mathbb{R}^6$ and we take

$$\mathbf{i} = \{2, 5, 6\},$$

$$\setminus \mathbf{i} = \{1, 3, 4\},$$

and

$$\nu_i = \lambda$$

then we have

$$\frac{d\pi}{d\lambda^6}(x_1, x_2, x_3, x_4, x_5, x_6) \stackrel{\nu^I}{=} \frac{d(\varpi_{\mathbf{i}})_*\pi}{d\lambda^3}(x_2, x_5, x_6) \frac{d\pi^{\varpi_{\mathbf{i}}}}{d\lambda^3}(x_1, x_3, x_4 \mid x_2, x_5, x_6)$$

This allows us to for example compute conditional probability densities from the evaluations of joint and marginal probability density functions,

$$\frac{d\pi^{\varpi_{\mathbf{i}}}}{d\lambda^3}(x_1, x_3, x_4 \mid x_2, x_5, x_6) \stackrel{\nu^I}{=} \frac{\frac{d\pi}{d\lambda^6}(x_1, x_2, x_3, x_4, x_5, x_6)}{\frac{d(\varpi_{\mathbf{i}})_*\pi}{d\lambda^3}(x_2, x_5, x_6)}.$$

Similarly

$$\begin{aligned}
\frac{d(\varpi_i)_*\pi}{d\lambda^3}(x_2, x_5, x_6) &\stackrel{\nu}{=} \int (\lambda^6)^{\varpi_i, \lambda^3}(\mathrm{d}x_{1:6} \mid x_2, x_5, x_6) \frac{d\pi}{d\lambda^6}(x_1, x_2, x_3, x_4, x_5, x_6) \\
&\stackrel{\nu}{=} \int \lambda^3(x_1, x_3, x_4) \frac{d\pi}{d\lambda^6}(x_1, x_2, x_3, x_4, x_5, x_6) \\
&\stackrel{\nu}{=} \int \mathrm{d}x_1 \mathrm{d}x_3 \mathrm{d}x_4 \frac{d\pi}{d\lambda^6}(x_1, x_2, x_3, x_4, x_5, x_6).
\end{aligned}$$

Given a joint probability density function we can use this result to compute the marginal probability density function, assuming we can evaluate the integral analytically, and then the product rule to compute the conditional probability density function. At the same time any marginal probability density function paired with a conditional probability density function defines a joint probability density function.

Addition example.

5 Convolutions

Probability space $(X, \mathcal{X}, \pi)$.

We can construct the second-order power space $X^2 = X \times X$ with projection functions ϖ_1 and ϖ_2 that return the first and second identical components, respectively. Any conditional probability distribution over X^2 with respect to ϖ_1 lifts π to a joint probability distribution

$$\pi(\mathrm{d}x_1, \mathrm{d}x_2) = \pi(\mathrm{d}x_1 \mid x_2) \pi(\mathrm{d}x_2).$$

By construction pushing this joint probability distribution forward along ϖ_1 returns the original probability distribution,

$$(\varpi_1)_*\pi(\mathrm{d}x_2) = \pi(\mathrm{d}x_2).$$

Pushing it forward along the second projection function ϖ_2 , however, generally gives a completely different probability distribution over X .

This process of lifting and then projecting to construct new probability distributions is known as **convolution**.

Comparison to additive pushforward.